STRICT HOT SPOTS AND BOUNDARY CRITICAL POINTS ON DOMAINS CLOSE TO A BALL

ABSTRACT. Fix a dimension $n \geq 2$ and a Hölder exponent $0 < \alpha < 1$. We prove that every domain whose boundary is a sufficiently small $C^{2,\alpha}$ radial perturbation of the unit sphere has the following property: every nonzero first positive Neumann eigenfunction has no interior critical point and has exactly two critical points on the closure, namely its unique boundary maximum and minimum. Both are nondegenerate critical points of the boundary restriction. Uniformly over these domains and eigenfunctions, the magnitude of the gradient is bounded below by a positive constant times the distance to the boundary times the supremum norm of the eigenfunction. The proof first establishes the complete structure of the first eigenspace of the ball, then proves uniform C^2 compactness under domain perturbations, and finally uses the Neumann condition and a nonzero second normal derivative to exclude critical points approaching the boundary. This argument accommodates arbitrary eigenvalue multiplicity and splitting. A general stability criterion is proved along the way. The near-ball theorem is independent of recent results on the planar hot spots conjecture.

Draft status. This candidate proof was obtained by our team on October 17, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. INTRODUCTION

1.1. **The problem and the current research status.** Let $\Omega \subset \mathbb{R}^n$ be a bounded connected domain with sufficiently regular boundary. Number its Neumann eigenvalues, with multiplicity, as

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \leq \mu_2(\Omega) \leq \cdots .$$

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The notation $E_1(\Omega)$ will mean the real eigenspace associated with the first *positive* eigenvalue $\mu_1(\Omega)$. Thus a nonzero $u \in E_1(\Omega)$ solves

$$-\Delta u = \mu_1(\Omega) u \quad \text{in } \Omega, \qquad \partial_\nu u = 0 \quad \text{on } \partial\Omega, \tag{1.1}$$

where ν is the outward unit normal. Integrating the equation shows that $\int_\Omega u = 0$.

Rauch's hot spots conjecture arose from the large-time behavior of the heat equation with an insulated boundary [11]. Its spectral form asks whether the maximum and minimum of a first positive Neumann eigenfunction occur on the boundary. We call the conclusion *strict* when

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \qquad \text{for every } x \in \Omega. \tag{1.2}$$

The absence of interior critical points, $\nabla u \neq 0$ in Ω, is a stronger assertion. Statements in this paper apply to every nonzero member of $E_1(\Omega)$, including when this eigenspace has dimension greater than one.

The unrestricted planar conjecture is false for domains with holes; Burdzy and Werner constructed a counterexample in [4]. Positive results have long been available under additional geometric or symmetry assumptions; see, for example, Bañuelos and Burdzy [2]. The literature status recorded here is as of October 17, 2026. OpenAI's September 2026 preprint [10] asserts that every first positive Neumann eigenfunction on a smooth bounded simply connected planar domain has no interior critical point. Accepting that result, the smooth simply connected planar case, and hence the smooth convex planar case, is settled in this strong form. The proofs in the present paper do not use that preprint.

There is a particularly close earlier result for the geometry studied here. Miyamoto [8] proved a criterion involving squared diameter divided by area for convex planar domains, which covers all sufficiently small smooth C^2 perturbations of the disk. His argument also excludes interior critical points. Consequently, the planar near-disk conclusion already has an antecedent predating the recent planar developments. In higher dimensions, Kennedy and Rohleder [7] established strong monotonicity conclusions for every first eigenfunction on suitable lip domains. They also obtained results for eigenfunctions in specified symmetry sectors on domains with coordinate symmetries. These hypotheses do not cover arbitrary small perturbations of a sphere without additional geometric structure.

Convexity alone does not imply the conjecture in every dimension. De Dios Pont [5] constructed smooth convex domains with interior hot spots in all sufficiently high dimensions. More recently, Atar and Burdzy [1] constructed a bounded Lipschitz starlike domain in $\mathbb{R}^3$ for which a first positive Neumann eigenfunction has a unique global maximum at an interior point. The radial boundary pieces in their construction show that this domain is not convex. The case of an arbitrary smooth convex domain in dimensions three and four remains unresolved to our knowledge. They also show why a local theorem around the ball must be distinguished from a theorem for all convex domains.

The purpose of this paper is to give a complete perturbative proof in every fixed dimension, with explicit attention to the multiple first eigenvalue of the ball. We also record a uniform lower bound for the gradient and the exact number of boundary critical points. No claim of priority for the qualitative near-ball principle is made. The proof uses variational spectral theory, elementary ordinary differential equations, and boundary Schauder estimates; it does not require a dimension-specific kernel representation.

1.2. **Main results.** Fix $0 < \alpha < 1$, and equip $C^{2,\alpha}(\mathbb{S}^{n-1})$ with any fixed standard Hölder norm. For $\rho \in C^{2,\alpha}(\mathbb{S}^{n-1})$ with $1 + \rho > 0$, set

$$\Omega_\rho = \{0\} \cup \{r\theta : \theta \in \mathbb{S}^{n-1},\ 0 < r < 1 + \rho(\theta)\}. \tag{1.3}$$

The function $\rho = 0$ gives the unit ball itself. Since the outward normal to the unit sphere is radial, this is also the usual normal-graph description of a domain close to the ball. We write $\|u\|_{\infty,\Omega} = \sup_{\overline{\Omega}} |u|$ for a continuous function on $\overline{\Omega}$.

Theorem 1.1 (Near-ball theorem)**.** *For every $n \geq 2$ and $0 < \alpha < 1$, there are constants $\varepsilon = \varepsilon(n,\alpha) > 0$ and $c = c(n,\alpha) > 0$ with the following property. If $\rho \in C^{2,\alpha}(\mathbb{S}^{n-1})$ satisfies*

$$\|\rho\|_{C^{2,\alpha}(\mathbb{S}^{n-1})} < \varepsilon,$$

then Ω_ρ is strictly convex, and every nonzero $u \in E_1(\Omega_\rho)$ has the following properties.

(i) *For every $x \in \Omega_\rho$,*

$$|\nabla u(x)| \geq c \operatorname{dist}(x, \partial\Omega_\rho)\, \|u\|_{\infty,\Omega_\rho}. \tag{1.4}$$

In particular, u has no interior critical point.

(ii) *The zero set of ∇u in $\overline{\Omega_\rho}$ consists of exactly two points p_- and p_+, both on the boundary. They are the unique global minimum and maximum of u, respectively. As critical points of $u|_{\partial\Omega_\rho}$, both are nondegenerate.*

(iii) *The strict hot spots inequalities hold:*

$$u(p_-) < u(x) < u(p_+) \qquad (x \in \Omega_\rho).$$

No simplicity assumption on $\mu_1(\Omega_\rho)$ is imposed.

Here a boundary critical point means a critical point of the restriction to the boundary. The Neumann condition makes this equivalent to the vanishing of the full Euclidean gradient at that boundary point. Nondegeneracy means that the Hessian of the restriction, as a quadratic form on the tangent space, is nonsingular. The maximum and minimum in Theorem 1.1 have negative and positive definite boundary Hessians, respectively.

The theorem includes smooth domains and the exact ball in every dimension under consideration, in particular in $\mathbb{R}^3$ and $\mathbb{R}^4$. The constants are uniform over all admissible perturbations and all normalized first eigenfunctions. The size of the admissible neighborhood is qualitative; we do not give a numerical value of ε. Translation and dilation give the corresponding statement near any ball; the precise scaling is recorded in Corollary 4.3.

A second result, Theorem 3.4, isolates the mechanism behind the perturbation argument. For a general reference domain, absence of interior critical points for all first eigenfunctions is stable if every boundary critical point has a nonzero second normal derivative. This criterion also yields the distance factor in (1.4). It does not require convexity of the reference domain.

1.3. **Outline of the proof.** The proof is organized in dependency order. In Section 2, spherical harmonics and a variational comparison identify the entire first eigenspace of the ball. Every normalized member is a rotation of one eigenfunction with a strictly increasing radial profile. Its only critical points on the closed ball are two antipodal boundary points, where the second normal derivatives are nonzero.

Section 3 pulls the Neumann problem back to a fixed domain. The min–max principle gives eigenvalue continuity, and a fixed-domain Schauder estimate gives uniform $C^{2,\alpha}$ bounds. Consequently, arbitrary normalized first eigenfunctions have subsequences converging in C^2. This compactness of the whole normalized eigenspace avoids any selection of eigenfunction branches. To rule out critical points near the boundary, we integrate the second derivative along the segment from a nearest boundary point, using the Neumann condition at its endpoint.

Finally, Section 4 constructs the required diffeomorphisms for radial graphs and applies the stability criterion to the ball. A separate C^2 argument for the boundary restrictions, which are close to height functions on the sphere, proves that the maximum and minimum are the only boundary critical points.

2. The first eigenspace of the ball

Throughout this section $B = B_1(0) \subset \mathbb{R}^n$, with $n \geq 2$. We first identify its first positive Neumann eigenspace without appealing to an ordering theorem for Bessel zeros.

Proposition 2.1 (Spectral structure of the ball)**.** *Put $\mu_B = \mu_1(B)$. There is a unique regular radial profile R, normalized by $R(1) = 1$, such that*

$$R'' + \frac{n-1}{r}R' + \left(\mu_B - \frac{n-1}{r^2}\right)R = 0 \quad (0 < r < 1), \qquad R'(1) = 0. \tag{2.1}$$

It is positive on $(0, 1]$, and

$$E_1(B) = \{U_a : a \in \mathbb{R}^n\}, \qquad U_a(r\theta) = R(r)(a \cdot \theta), \quad U_a(0) = 0. \tag{2.2}$$

In particular, $\dim E_1(B) = n$.

Proof. The spherical harmonics of degree ℓ have angular eigenvalue $a_\ell = \ell(\ell + n - 2)$. For an $L^2(\mathbb{S}^{n-1})$-normalized spherical harmonic Y_ℓ, the Rayleigh quotient of $F(r)Y_\ell(\theta)$ is

$$Q_\ell(F) = \frac{\displaystyle\int_0^1 (F'(r)^2 r^{n-1} + a_\ell F(r)^2 r^{n-3})\,\mathrm{d}r}{\displaystyle\int_0^1 F(r)^2 r^{n-1}\,\mathrm{d}r}. \tag{2.3}$$

For every $\ell \geq 1$, its radial form domain is

$$X = \left\{F \in H^1_{\mathrm{loc}}(0,1) : \int_0^1 (F'^2 r^{n-1} + F^2 r^{n-3})\,\mathrm{d}r < \infty\right\}.$$

Indeed, $a_\ell > 0$, and the displayed integral is equivalent to the squared $H^1(B)$ norm in each fixed angular sector: the additional L^2 term is controlled by $r^{n-1} \leq r^{n-3}$. Cutoffs near the origin are dense in this form domain: the cutoff-derivative error is bounded by a constant times $\int_0^{2\delta} F^2 r^{n-3}\,\mathrm{d}r$, which tends to zero as $\delta \downarrow 0$. Thus there is no additional trace condition at 0 hidden in this description. The embedding of X into $L^2((0,1), r^{n-1}\,\mathrm{d}r)$ is compact, by the compact embedding of $H^1(B)$ into $L^2(B)$.

Let $m = \min_{X\setminus\{0\}} Q_1$. A minimizing sequence normalized in $L^2((0,1), r^{n-1}\,\mathrm{d}r)$ is bounded in X, so compactness and weak lower semicontinuity give a minimizer. It may be chosen nonnegative, since replacing its profile by its absolute value preserves the quotient. The Euler equation is (2.1) with m in place of μ_B, with the natural condition $R'(1) = 0$. To verify the equation against an arbitrary $\varphi \in H^1(B)$, fix a normalized degree-one harmonic Y_1 and put

$$G(r) = \int_{\mathbb{S}^{n-1}} \varphi(r\theta)Y_1(\theta)\,\mathrm{d}\theta.$$

Cauchy–Schwarz and $-\Delta_{\mathbb{S}^{n-1}}Y_1 = a_1 Y_1$ give

$$|G'(r)|^2 \leq \int_{\mathbb{S}^{n-1}} |\partial_r \varphi(r\theta)|^2\,\mathrm{d}\theta, \qquad a_1|G(r)|^2 \leq \int_{\mathbb{S}^{n-1}} |\nabla_{\mathbb{S}^{n-1}}(\varphi(r\theta))|^2\,\mathrm{d}\theta.$$

Integration with the polar weights shows that $G \in X$. Angular orthogonality then makes the Euler identity tested with G exactly

$$\int_B \nabla(RY_1) \cdot \nabla\varphi\,\mathrm{d}x = m\int_B (RY_1)\varphi\,\mathrm{d}x.$$

Thus the resulting function is a weak Neumann eigenfunction on the whole ball. Moreover $m > 0$, since $Q_1(F) \geq n - 1$. Elliptic regularity supplies its smoothness, including at the origin. The radial profile is strictly positive on $(0, 1]$: an interior zero would also have zero first derivative, and a zero at 1 would have the same consequence by the Neumann condition. In either case, uniqueness for the ordinary differential equation would force the profile to vanish identically.

For $\ell \geq 2$, the common form domain and $r^{n-3} \geq r^{n-1}$ give

$$Q_\ell(F) = Q_1(F) + (a_\ell - a_1)\frac{\int_0^1 F^2 r^{n-3}\,\mathrm{d}r}{\int_0^1 F^2 r^{n-1}\,\mathrm{d}r} \geq m + a_\ell - a_1 > m. \tag{2.4}$$

It remains to exclude positive eigenvalues from the radial sector $\ell = 0$. Let

$$d_1 = \min_{\substack{F \in X \setminus \{0\} \\ F(1)=0}} Q_1(F).$$

The trace at 1 is continuous because $X \subset H^1(1/2, 1)$, so its zero set is closed in X and the same compactness argument attains this minimum. We have $d_1 > m$. Otherwise a Dirichlet minimizer would also minimize the unrestricted Neumann quotient, and hence would satisfy both $F(1) = 0$ and $F'(1) = 0$, contrary to uniqueness for the ordinary differential equation.

If $V(r)$ is a nonconstant radial Neumann eigenfunction with eigenvalue $\lambda > 0$, differentiating its radial equation shows that $S = V'$ is a nonzero regular degree-one radial solution with eigenvalue λ and $S(1) = 0$. Smoothness of the radial function at the origin gives $S(r) = O(r)$ and $S'(r) = O(1)$ there, so $S \in X$. Multiplying its equation by Sr^{n-1} and integrating by parts gives

$$\int_0^1 (S'^2 r^{n-1} + (n-1)S^2 r^{n-3}) \, \mathrm{d}r = \lambda \int_0^1 S^2 r^{n-1} \, \mathrm{d}r.$$

The boundary term at 1 vanishes because $S(1) = 0$, and the one at 0 vanishes because $r^{n-1}SS' = O(r^n)$. Thus $Q_1(S) = \lambda \geq d_1 > m$.

Each nonzero spherical-harmonic projection of a Neumann eigenfunction satisfies the equation and boundary condition in its angular sector, with the same eigenvalue. Completeness of the spherical harmonics now implies that m is the first positive eigenvalue of the ball, and that its eigenspace lies entirely in degree one. At this eigenvalue a radial solution satisfying $R'(1) = 0$ is determined up to a scalar by its value at 1, which cannot vanish. The degree-one spherical harmonics are exactly the functions $a \cdot \theta$. This proves (2.2) and the asserted normalization and dimension. □

Lemma 2.2 (Monotonicity and endpoint behavior)**.** *The profile in Proposition 2.1 satisfies*

$$n - 1 < \mu_B < n + 2, \qquad R'(r) > 0 \quad (0 \leq r < 1), \qquad R''(1) = -(\mu_B - n + 1) < 0. \tag{2.5}$$

Moreover, $R(r) = ar + O(r^3)$ *near* 0*, with* $a > 0$*, and*

$$\frac{R(r)}{r} > R'(r) > 0 \qquad (0 < r < 1). \tag{2.6}$$

Proof. The trial profile $F(r) = r$ in Q_1 gives $\mu_B \leq n + 2$. Equality would make this trial profile a minimizer, whereas its derivative at 1 is nonzero. Thus $\mu_B < n + 2$.

For a fixed unit vector e, the function $t \mapsto U_e(te)$ is smooth and odd. Division by t therefore gives a smooth even extension of $f(r) = R(r)/r$ across zero. In particular, $R(r) = ar + O(r^3)$ for some $a \geq 0$. Set $H(r) = r^{n-1}R'(r)$. Equation (2.1) yields

$$H'(r) = r^{n-3}(n - 1 - \mu_B r^2)R(r), \qquad H(0) = H(1) = 0. \tag{2.7}$$

If $\mu_B \leq n - 1$, this derivative would be strictly positive for $0 < r < 1$, a contradiction. Hence $\mu_B > n - 1$. The derivative in (2.7) is positive before $r_* = [(n-1)/\mu_B]^{1/2}$ and negative after r_*. Since both endpoint values of H are zero, $H > 0$ in $(0, 1)$, proving $R' > 0$ there.

Substitution of $R = rf$ into the equation gives

$$(r^{n+1}f')' = -\mu_B r^{n+1} f, \qquad f'(r) = -\mu_B r^{-n-1} \int_0^r t^{n+1} f(t) \, \mathrm{d}t. \tag{2.8}$$

The lower-end boundary term vanishes because f is smooth at zero. Since $f(r) = R(r)/r > 0$ on $(0, 1]$, the integral formula gives $f' < 0$ there. Consequently $a = f(0) \geq f(r) > 0$ for every $0 < r \leq 1$, so $R'(0) = a > 0$. Together with the flux argument, the identity $R' = f + rf'$ proves (2.6). Finally, evaluating (2.1) at 1 gives the last identity in (2.5). □

Proposition 2.3 (Critical points and boundary nondegeneracy on the ball)**.** *For a unit vector $e \in \mathbb{S}^{n-1}$, let U_e be as in* (2.2)*. Then $\|U_e\|_{\infty,B} = 1$ and*

$$\{x \in \overline{B} : \nabla U_e(x) = 0\} = \{-e, e\}. \tag{2.9}$$

If $\beta_n = \mu_B - n + 1 > 0$, the ambient Hessians at these points are

$$D^2U_e(e) = -(I - e \otimes e) - \beta_n e \otimes e, \tag{2.10}$$

$$D^2U_e(-e) = (I - e \otimes e) + \beta_n e \otimes e. \tag{2.11}$$

There is also a constant $m_n > 0$ such that

$$|\nabla U_e(x)| \geq m_n(1 - |x|) \qquad (x \in B) \tag{2.12}$$

for every unit vector e.

Proof. For $x = r\theta \neq 0$, radial and tangential differentiation gives

$$\nabla U_e(r\theta) = R'(r)(e \cdot \theta)\theta + \frac{R(r)}{r}(e - (e \cdot \theta)\theta). \tag{2.13}$$

The two summands are orthogonal, so

$$|\nabla U_e(r\theta)|^2 = R'(r)^2(e \cdot \theta)^2 + \left(\frac{R(r)}{r}\right)^2 (1 - (e \cdot \theta)^2). \tag{2.14}$$

Both coefficients are positive when $0 < r < 1$, and at the origin $\nabla U_e(0) = ae \neq 0$. On the sphere the gradient is $e - (e \cdot \theta)\theta$, whose only zeros are $\pm e$. Since $R(r) < 1$ for $r < 1$ and $R(1) = 1$, the supremum norm is one. The global maximum 1 and minimum -1 are attained uniquely at e and $-e$, respectively.

To compute the Hessian, write $U_e(x) = f(|x|)(e \cdot x)$ as above. At $r = 1$ one has $f'(1) = R'(1) - R(1) = -1$. Hence the second derivative at e in every unit tangential direction is -1, the second radial derivative is $R''(1) = -\beta_n$, and the mixed derivatives vanish. This gives (2.10). Since U_e is odd, its Hessian is odd, giving (2.11). At these critical points the Hessian of the boundary restriction is the tangential restriction of the ambient Hessian, since the full gradient vanishes. Its tangential eigenvalues are therefore -1 at e and 1 at $-e$; in particular both boundary critical points are nondegenerate. The second normal derivatives are $-\beta_n$ and β_n, respectively.

The quotient $R'(r)/(1 - r)$ extends continuously and positively to $[0, 1]$, with endpoint values a at 0 and β_n at 1. Therefore

$$m_n = \min_{0 \leq r \leq 1} \frac{R'(r)}{1 - r} > 0,$$

where the quotient at 1 is interpreted by continuity. Equations (2.6) and (2.14) imply $|\nabla U_e(r\theta)| \geq R'(r) \geq m_n(1 - r)$. The estimate at the origin follows by continuity. □

3. Compactness and a stability criterion

In this section $D \subset \mathbb{R}^n$ is a fixed bounded connected $C^{2,\alpha}$ domain. A $C^{2,\alpha}$ diffeomorphism of closures is understood to have a $C^{2,\alpha}$ inverse and to map the boundary onto the boundary. All the maps below are close to the identity in the indicated norm on $\overline{D}$.

3.1. **Spectral and eigenfunction compactness.** Stability of Neumann eigenvalues and eigenfunctions has been studied in substantially more general settings; see Bañuelos and Pang [3]. We give the fixed-domain argument needed here, including the uniform estimates that allow eigenvalue multiplicity.

Lemma 3.1 (Compactness of arbitrary first eigenfunctions)**.** *Let $F_j : \overline{D} \to \overline{\Omega_j}$ be $C^{2,\alpha}$ diffeomorphisms satisfying*

$$F_j \longrightarrow \mathrm{id} \quad \text{in } C^{2,\alpha}(\overline{D}).$$

Then $\mu_1(\Omega_j) \to \mu_1(D)$. If $u_j \in E_1(\Omega_j)$ and $\|u_j\|_{\infty,\Omega_j} = 1$, a subsequence satisfies

$$v_j := u_j \circ F_j \longrightarrow v \quad \text{in } C^2(\overline{D}), \qquad v \in E_1(D), \quad \|v\|_{\infty,D} = 1. \tag{3.1}$$

The choice of u_j is arbitrary at every index.

Proof. Step 1: pullback of the variational problem. For all sufficiently large j, put

$$J_j = \det DF_j > 0, \qquad A_j = J_j DF_j^{-1} DF_j^{-T}.$$

Then $J_j \to 1$ and $A_j \to I$ in $C^{1,\alpha}(\overline{D})$. Writing $\lambda_j = \mu_1(\Omega_j)$, change of variables in the weak equation gives

$$\int_D A_j \nabla v_j \cdot \nabla \phi \, \mathrm{d}x = \lambda_j \int_D J_j v_j \phi \, \mathrm{d}x \qquad (\phi \in H^1(D)). \tag{3.2}$$

In particular $\int_D J_j v_j = 0$. The corresponding differential and boundary equations are

$$\mathrm{div}(A_j \nabla v_j) + \lambda_j J_j v_j = 0 \quad \text{in } D, \qquad (A_j \nabla v_j) \cdot \nu_D = 0 \quad \text{on } \partial D. \tag{3.3}$$

Step 2: eigenvalue continuity. For numbers $\delta_j \downarrow 0$, uniform convergence of the coefficients gives, for every nonzero $w \in H^1(D)$,

$$\frac{1-\delta_j}{1+\delta_j} \frac{\int_D |\nabla w|^2}{\int_D w^2} \le \frac{\int_D A_j \nabla w \cdot \nabla w}{\int_D J_j w^2} \le \frac{1+\delta_j}{1-\delta_j} \frac{\int_D |\nabla w|^2}{\int_D w^2}. \tag{3.4}$$

The min–max characterization over $(k+1)$-dimensional subspaces of $H^1(D)$, including the constant eigenfunction at index 0, therefore gives $\mu_k(\Omega_j) \to \mu_k(D)$ for every fixed k. In particular, $\lambda_j \to \mu_1(D)$ and the numbers λ_j are bounded. This argument requires no simplicity assumption.

Step 3: a uniform boundary estimate. We use the fixed-domain Neumann Schauder estimate

$$\|w\|_{C^{2,\alpha}(\overline{D})} \le C_D \left(\|\Delta w\|_{C^{0,\alpha}(\overline{D})} + \|\partial_{\nu_D} w\|_{C^{1,\alpha}(\partial D)} + \|w\|_{C^0(\overline{D})} \right). \tag{3.5}$$

This a priori estimate is stated in Nardi [9, Theorem 3.2], with a reference to Gilbarg and Trudinger [6, Theorem 6.30]. The boundary-norm formulation follows by a bounded extension of $C^{1,\alpha}(\partial D)$ data on the fixed domain; see also [9, Remark 2.3]. The C^0 term is retained.

We justify regularity for every weak eigenfunction before applying this a priori estimate or using supremum normalization. Fix a bounded connected $C^{2,\alpha}$ domain Ω, and let $u \in H^1(\Omega)$ be a weak Neumann eigenfunction with eigenvalue $\lambda > 0$. The homogeneous Neumann resolvent of $-\Delta + 1$ maps $L^p(\Omega)$ to $W^{2,p}(\Omega)$ for every finite $p \ge 2$. Whenever $u \in L^p$, apply this result to $(-\Delta + 1)u = (\lambda + 1)u$; coercive weak uniqueness identifies the $W^{2,p}$ solution with u. Starting from $u \in L^2$, iterate this regularity and Sobolev embedding. When $2p < n$, the new integrability exponent is $np/(n-2p)$; when $2p = n$, every finite integrability exponent is available; and when $2p > n$, u is bounded. In the first case the reciprocal exponent decreases by $2/n$, so boundedness is reached after finitely many steps. A further resolvent application with $p > n$ gives $u \in W^{2,p}(\Omega) \subset C^{1,\gamma}(\overline{\Omega})$ for some $\gamma > 0$. In particular, $u \in C^{0,\alpha}(\overline{\Omega})$ for the prescribed $\alpha < 1$.

The mean-zero identity noted after (1.1) also follows for weak eigenfunctions by testing with 1. Thus the data $f = -\lambda u$ and $g = 0$ satisfy the Neumann compatibility condition. By [9, Theorem 3.1], there is a mean-zero $z \in C^{2,\alpha}(\overline{\Omega})$ with $\Delta z = -\lambda u$ and $\partial_\nu z = 0$. Subtracting its weak equation from that of u and testing with $u - z$ gives $\int_\Omega |\nabla(u-z)|^2 = 0$. Connectedness and the equal means imply $u = z$. This proves $C^{2,\alpha}$ regularity for every weak positive Neumann eigenfunction and, in particular, makes supremum normalization available for every nonzero member of $E_1(\Omega)$. Applying this argument on each Ω_j and composing with F_j shows that $v_j \in C^{2,\alpha}(\overline{D})$. No uniform estimate on the varying domains is used in this preliminary regularity argument; uniformity will follow from the estimate on the fixed domain D.

Rewrite (3.3) as

$$\Delta v_j = (I - A_j) : D^2 v_j - (\operatorname{div} A_j) \cdot \nabla v_j - \lambda_j J_j v_j, \tag{3.6}$$

$$\partial_{\nu_D} v_j = ((I - A_j)\nu_D) \cdot \nabla v_j. \tag{3.7}$$

Here the colon denotes contraction of matrices. Because $\nu_D \in C^{1,\alpha}$, the terms containing $A_j - I$ in (3.5) are bounded by

$$C\|A_j - I\|_{C^{1,\alpha}(\overline{D})}\|v_j\|_{C^{2,\alpha}(\overline{D})}.$$

They can be absorbed into the left-hand side for large j. The remaining eigenvalue term is bounded by $C\|v_j\|_{C^{0,\alpha}}$, since λ_j and J_j are uniformly bounded in the required norms. The interpolation inequality

$$\|w\|_{C^{0,\alpha}} \le \eta\|w\|_{C^{2,\alpha}} + C_\eta\|w\|_{C^0}$$

then permits a second absorption, with η fixed sufficiently small. We obtain

$$\|v_j\|_{C^{2,\alpha}(\overline{D})} \le C\|v_j\|_{C^0(\overline{D})} = C, \tag{3.8}$$

where C is independent of the index and of the chosen eigenfunction.

Step 4: identification of the limit. The compact embedding of $C^{2,\alpha}(\overline{D})$ into $C^2(\overline{D})$ gives a convergent subsequence. Passing to the limit in (3.2), using $A_j \to I$, $J_j \to 1$, and $\lambda_j \to \mu_1(D)$, gives

$$\int_D \nabla v \cdot \nabla\phi \,\mathrm{d}x = \mu_1(D)\int_D v\phi \,\mathrm{d}x \qquad (\phi \in H^1(D)).$$

Also $\int_D v = 0$, by passing to the limit in $\int_D J_j v_j = 0$. The supremum normalization passes to the limit, so v is nonzero and has norm one. Thus $v \in E_1(D)$, as asserted. □

Remark 3.2. The convergence in Lemma 3.1 may also be stated in $C^{2,\beta}$ for every $0 < \beta < \alpha$. A bounded set in $C^{2,\alpha}$ need not be compact in that same norm; no such convergence is asserted. A fixed positive Hölder exponent is used throughout the perturbation argument.

Corollary 3.3 (Uniform approximation of the normalized eigenspace)**.** *Let*

$$\mathcal{S}_D = \{v \in E_1(D) : \|v\|_{\infty,D} = 1\}.$$

This is a compact subset of $C^2(\overline{D})$*. As* $F \to \mathrm{id}$ *in* $C^{2,\alpha}(\overline{D})$*,*

$$\sup_{\substack{u \in E_1(F(D)) \\ \|u\|_{\infty,F(D)}=1}} \inf_{v \in \mathcal{S}_D} \|u \circ F - v\|_{C^2(\overline{D})} \longrightarrow 0. \tag{3.9}$$

Proof. Compactness of $\mathcal{S}_D$ follows by taking $F_j = \mathrm{id}$ in Lemma 3.1. If (3.9) failed, a sequence of maps and normalized eigenfunctions violating it would contradict Lemma 3.1. □

We shall also need convergence of the physical derivatives, rather than only derivatives of the pullbacks. The chain rule gives

$$(\nabla u_j) \circ F_j = DF_j^{-T} \nabla v_j, \tag{3.10}$$

$$D^2 v_j = DF_j^T [(D^2 u_j) \circ F_j] DF_j + \sum_{k=1}^{n} [(\partial_k u_j) \circ F_j] D^2 F_{j,k}. \tag{3.11}$$

Consequently, along the subsequence in Lemma 3.1,

$$(\nabla u_j) \circ F_j \to \nabla v, \qquad (D^2 u_j) \circ F_j \to D^2 v \quad \text{uniformly on } \overline{D}. \tag{3.12}$$

The outward normals satisfy

$$\nu_{\Omega_j}(F_j(s)) = \frac{DF_j(s)^{-T}\nu_D(s)}{|DF_j(s)^{-T}\nu_D(s)|} \longrightarrow \nu_D(s) \quad \text{uniformly for } s \in \partial D. \tag{3.13}$$

3.2. **Excluding critical points approaching the boundary.**

Theorem 3.4 (A boundary nondegeneracy criterion)**.** *Suppose that every nonzero $v \in E_1(D)$ satisfies*

(i) $\nabla v \neq 0$ *in* D*;*

(ii) *if* $s \in \partial D$ *and* $\nabla v(s) = 0$*, then*

$$D^2 v(s)[\nu_D(s), \nu_D(s)] \neq 0. \tag{3.14}$$

There exist $\varepsilon_D, c_D > 0$ *such that every* $C^{2,\alpha}$ *diffeomorphism* F *with* $\|F - \mathrm{id}\|_{C^{2,\alpha}(\overline{D})} < \varepsilon_D$ *and every* $u \in E_1(\Omega)$*, where* $\Omega = F(D)$*, satisfy*

$$|\nabla u(x)| \geq c_D \operatorname{dist}(x, \partial\Omega) \|u\|_{\infty,\Omega} \qquad (x \in \Omega). \tag{3.15}$$

Proof. The estimate is immediate for $u = 0$, and for nonzero u it is invariant under multiplication by a nonzero scalar. If the assertion were false, for every positive integer j we could choose F_j with $\|F_j - \mathrm{id}\|_{C^{2,\alpha}} < 1/j$, a first eigenfunction u_j normalized by $\|u_j\|_{\infty,\Omega_j} = 1$, and $x_j \in \Omega_j = F_j(D)$ such that, with $d_j = \operatorname{dist}(x_j, \partial\Omega_j) > 0$,

$$\frac{|\nabla u_j(x_j)|}{d_j} < \frac{1}{j}. \tag{3.16}$$

Apply Lemma 3.1 and pass to a subsequence for which $u_j \circ F_j \to v \in \mathcal{S}_D$ in C^2 and $F_j^{-1}(x_j) \to x \in \overline{D}$. The domains are uniformly bounded, so the numbers d_j are bounded. Equations (3.16) and (3.12) give $\nabla v(x) = 0$. Assumption (i) forces $x \in \partial D$, and $d_j \leq |x_j - F_j(x)| \to 0$.

Choose a nearest boundary point $b_j \in \partial\Omega_j$ to x_j, and set $\nu_j = \nu_{\Omega_j}(b_j)$. Minimizing the squared distance along the C^1 boundary gives $(x_j - b_j) \cdot \tau = 0$ for every $\tau \in T_{b_j}\partial\Omega_j$. Thus $x_j - b_j = \pm d_j \nu_j$, and in fact

$$x_j = b_j - d_j \nu_j. \tag{3.17}$$

The sign is inward because the open segment from b_j to x_j lies in the open ball $B(x_j, d_j) \subset \Omega_j$. This observation also shows that no uniqueness of the nearest boundary point is required. Since $x_j \to x$ and $|b_j - x_j| = d_j \to 0$, we have $b_j \to x$. Writing $s_j = F_j^{-1}(b_j) \in \partial D$, the bound $|s_j - b_j| \leq \|F_j - \mathrm{id}\|_{C^0(\overline{D})}$ gives $s_j \to x$. Equation (3.13) and continuity of ν_D therefore imply $\nu_j \to \nu_D(x)$.

Put

$$a = D^2 v(x)[\nu_D(x), \nu_D(x)] \neq 0.$$

Every point $b_j - t\nu_j$, $0 \leq t \leq d_j$, approaches x uniformly in t. Its preimage under F_j does so as well. By (3.12) and continuity of $D^2 v$,

$$\sup_{0 \leq t \leq d_j} \left| D^2 u_j(b_j - t\nu_j)[\nu_j, \nu_j] - a \right| \longrightarrow 0. \tag{3.18}$$

For large j, these normal second derivatives therefore have the sign of a and magnitude at least $|a|/2$ along the whole segment. Holding the vector ν_j fixed and using the Neumann condition at b_j, the fundamental theorem of calculus gives

$$\nabla u_j(x_j) \cdot \nu_j = -\int_0^{d_j} D^2 u_j(b_j - t\nu_j)[\nu_j, \nu_j] \, \mathrm{d}t. \tag{3.19}$$

It follows that $|\nabla u_j(x_j)| \geq |a| d_j/2$, contradicting (3.16). □

The distance in (3.15) is essential to the form of the statement: the gradient of a Neumann eigenfunction vanishes at a boundary maximum or minimum. The estimate controls its decay toward the boundary while still excluding every interior critical point.

4. Application to domains close to a ball

4.1. **Radial graphs and their geometry.**

Lemma 4.1 (A fixed-domain parametrization)**.** *For all sufficiently small $\|\rho\|_{C^{2,\alpha}(\mathbb{S}^{n-1})}$, there is a $C^{2,\alpha}$ diffeomorphism $F_\rho : \overline{B} \to \overline{\Omega_\rho}$ such that*

$$F_\rho(\theta) = (1+\rho(\theta))\theta \quad (\theta \in \mathbb{S}^{n-1}), \qquad \|F_\rho - \mathrm{id}\|_{C^{2,\alpha}(\overline{B})} \le C\|\rho\|_{C^{2,\alpha}(\mathbb{S}^{n-1})}. \tag{4.1}$$

For a sufficiently small C^2 norm of ρ, the domain Ω_ρ is strictly convex.

Proof. Choose a fixed smooth function $\chi : [0,1] \to [0,1]$ which is zero near $[0,1/3]$ and one near $[2/3,1]$. Define

$$F_\rho(r\theta) = (r + \chi(r)\rho(\theta))\theta, \qquad F_\rho(0) = 0. \tag{4.2}$$

This is the identity near the origin. For small $\|\rho\|_{C^0}$, the radial derivative $1+\chi'(r)\rho(\theta)$ is positive, and each ray is mapped bijectively onto the ray with terminal radius $1+\rho(\theta)$. More explicitly, for $r > 0$,

$$\det DF_\rho(r\theta) = (1+\chi'(r)\rho(\theta))\left(1 + \frac{\chi(r)\rho(\theta)}{r}\right)^{n-1} > 0.$$

The second factor is positive for small $\|\rho\|_{C^0}$ because $r \ge 1/3$ wherever $\chi(r) \ne 0$. Thus the map is globally bijective and locally nonsingular. Since $\chi = 1$ near $r = 1$, extend it by one slightly beyond 1. The same formula then defines a $C^{2,\alpha}$ map on a neighborhood of the boundary with nonzero Jacobian. The inverse function theorem therefore gives a $C^{2,\alpha}$ inverse up to the boundary as well as in the interior. Hence F_ρ is a diffeomorphism of closures. On a fixed annulus containing the support of χ, the radial projection $x \mapsto x/|x|$ and all required derivatives are bounded. Composition with this projection and multiplication by the fixed smooth cutoff are bounded operations in $C^{2,\alpha}$. Together with $F_\rho = \mathrm{id}$ near zero, this proves (4.1).

For convexity, define

$$G_\rho(x) = \frac{|x|^2}{(1+\rho(x/|x|))^2} \quad (x \ne 0), \qquad G_\rho(0) = 0.$$

The function is homogeneous of degree two, and its Hessian away from the origin is homogeneous of degree zero. Differentiating $|x|^2[(1+\rho(x/|x|))^{-2} - 1]$ on the unit sphere gives

$$\sup_{|x|=1} \|D^2 G_\rho(x) - 2I\|_{\mathrm{op}} \le C_n \|\rho\|_{C^2(\mathbb{S}^{n-1})}$$

when $\|\rho\|_{C^2(\mathbb{S}^{n-1})}$ is small. Thus, for a sufficiently small C^2 norm, $D^2 G_\rho \ge I$ everywhere away from zero. Also G_ρ is C^1 at the origin, with gradient zero there: homogeneity and boundedness on the sphere give $G_\rho(x) = O(|x|^2)$ and $\nabla G_\rho(x) = O(|x|)$. For distinct $x, y \in \mathbb{R}^n$, the restriction $\gamma(t) = G_\rho((1-t)x+ty)$ is continuously differentiable and satisfies $\gamma''(t) \ge |y-x|^2$ except possibly at the single parameter where the segment passes through zero. Continuity of γ' permits integration across that parameter, giving

$$G_\rho((1-t)x+ty) \le (1-t)G_\rho(x) + tG_\rho(y) - \frac{1}{2}t(1-t)|x-y|^2 \qquad (0 \le t \le 1).$$

For distinct $x, y \in \overline{\Omega_\rho}$ and $0 < t < 1$, the right-hand side is strictly less than one. Thus the sublevel set $\Omega_\rho = \{G_\rho < 1\}$ is strictly convex. On the level set $G_\rho = 1$ the gradient is nonzero, since $x \cdot \nabla G_\rho = 2G_\rho$. With the outward unit normal $\nu = \nabla G_\rho / |\nabla G_\rho|$, every nonzero tangent vector τ satisfies

$$\langle D_\tau \nu, \tau \rangle = \frac{D^2 G_\rho[\tau,\tau]}{|\nabla G_\rho|} > 0.$$

This also proves positivity of the boundary second fundamental form in the displayed outward-normal convention. □

4.2. **Critical points of functions close to a spherical height.** Let g_S be the standard metric on $\mathbb{S}^{n-1}$, and let ∇_S and Hess_S denote its gradient and Hessian. We use a C^2 norm on the sphere controlling the function and these two derivatives.

Lemma 4.2 (Stability of the two boundary critical points)**.** *There is $\eta > 0$, depending only on the dimension and the chosen C^2 norm, such that the following holds. If $e \in \mathbb{S}^{n-1}$ and $h \in C^2(\mathbb{S}^{n-1})$ satisfy*

$$\|h - h_e\|_{C^2(\mathbb{S}^{n-1})} < \eta, \qquad h_e(\theta) = e \cdot \theta,$$

then h has exactly two critical points. One is its unique maximum and has negative definite Hessian, and the other is its unique minimum and has positive definite Hessian.

Proof. The height function satisfies

$$|\nabla_S h_e|^2 = 1 - h_e^2, \qquad \mathrm{Hess}_S\, h_e = -h_e g_S. \tag{4.3}$$

Consider the geodesically convex caps

$$C_+ = \{\theta : h_e(\theta) > 3/4\}, \qquad C_- = \{\theta : h_e(\theta) < -3/4\}.$$

Each cap is a geodesic ball of radius $\arccos(3/4) < \pi/2$, so any two of its points are joined by a unique minimizing geodesic lying inside that cap; this includes the circle when $n = 2$. Outside these caps, $|\nabla_S h_e| \geq \sqrt{7}/4$. In C_+ the Hessian of h_e is bounded above by $-3g_S/4$, and in C_- it is bounded below by $3g_S/4$. Choose η so that the C^2 error controls the pointwise gradient error by $\sqrt{7}/8$ and the Hessian error in operator norm by $1/4$. Then $|\nabla_S h| \geq \sqrt{7}/8$ outside $C_+ \cup C_-$, and

$$\mathrm{Hess}_S\, h \leq -\tfrac{1}{2} g_S \text{ in } C_+, \qquad \mathrm{Hess}_S\, h \geq \tfrac{1}{2} g_S \text{ in } C_-.$$

In particular h is strictly geodesically concave in C_+ and strictly geodesically convex in C_-. Since the fixed C^2 norm controls the intrinsic derivatives independently of e, the same η works for every direction.

There is at most one critical point in either cap. Indeed, along the unit-speed minimizing geodesic joining two putative critical points in the same cap, the derivative of h would vanish at both endpoints, whereas its second derivative has a fixed strict sign. A maximum and a minimum exist by compactness of the sphere. They must lie in the caps, and the Hessian signs force the maximum into C_+ and the minimum into C_-. This gives exactly one critical point in each, with the asserted nondegeneracy and uniqueness. □

4.3. **Proof of the near-ball theorem.**

Proof of Theorem 1.1. First choose the perturbation size small enough for Lemma 4.1. This provides the parametrizations F_ρ and strict convexity.

Every nonzero member of $E_1(B)$ is a nonzero scalar multiple of U_e for some unit vector e, by Proposition 2.1. Proposition 2.3 shows that it has no interior critical point. At the only two boundary critical points, (2.10) and (2.11) show that its second normal derivative is nonzero. Thus the reference domain $D = B$ satisfies both hypotheses of Theorem 3.4. Equations (4.1) and (3.15) give (1.4) after decreasing ε if necessary. The constants depend only on n, α, and the fixed norm and cutoff conventions. In particular, no interior critical point can occur.

It remains to determine the boundary critical points. Normalize $\|u\|_{\infty,\Omega_\rho} = 1$, and put

$$v = u \circ F_\rho \quad \text{on } \overline{B}, \qquad h = v|_{\mathbb{S}^{n-1}}.$$

By Corollary 3.3, uniformly over all these normalized eigenfunctions, v is arbitrarily close in $C^2(\overline{B})$ to some member of $\mathcal{S}_B$ as $\|\rho\|_{C^{2,\alpha}} \to 0$. Here $\mathcal{S}_B = \{U_e : e \in \mathbb{S}^{n-1}\}$ by Propositions 2.1 and 2.3. Restriction to the sphere is continuous from $C^2(\overline{B})$ to $C^2(\mathbb{S}^{n-1})$, so h is arbitrarily close in C^2 to $h_e(\theta) = e \cdot \theta$ for some e. The direction e may depend on the domain and the eigenfunction; no continuous choice is needed.

Shrink ε once more so that Lemma 4.2 applies. It follows that h, and hence $u|_{\partial\Omega_\rho}$, has exactly two critical points, with the stated Hessian signs. Nondegeneracy at a critical point is preserved under a diffeomorphism of the underlying boundary. Indeed, writing $\Phi = F_\rho|_{\mathbb{S}^{n-1}}$ and $q = u|_{\partial\Omega_\rho}$, at a critical point θ of $h = q \circ \Phi$ the Hessians satisfy

$$\operatorname{Hess}_{\mathbb{S}^{n-1}} h(\theta)[\xi,\xi] = \operatorname{Hess}_{\partial\Omega_\rho} q(\Phi(\theta))[D\Phi(\theta)\xi, D\Phi(\theta)\xi].$$

There is no term involving second derivatives of Φ because $\mathrm{d}q = 0$ at that point. Since $D\Phi$ is an isomorphism of tangent spaces, the signs and nondegeneracy of these quadratic forms agree. The Neumann condition identifies these points with the boundary zeros of the full gradient. Since there are no interior zeros, these are the only zeros of ∇u on the closure.

The continuous nonconstant function u attains its extrema on the compact closure. An interior extremum would be an interior critical point. Both extrema therefore lie on the boundary and, by the uniqueness just proved, are precisely the two points p_- and p_+. No interior point can have either extremal value. This proves the strict inequalities and completes the proof. General amplitudes follow by rescaling u. □

Corollary 4.3 (Translation and scaling)**.** *Let $x_0 \in \mathbb{R}^n$, $R_0 > 0$, and $\Omega = x_0 + R_0\Omega_\rho$, where ρ satisfies the smallness assumption of Theorem 1.1. Every nonzero $u \in E_1(\Omega)$ has exactly the two boundary critical points described there and satisfies*

$$|\nabla u(x)| \geq \frac{c}{R_0^2} \operatorname{dist}(x, \partial\Omega)\|u\|_{\infty,\Omega} \qquad (x \in \Omega). \tag{4.4}$$

Proof. Apply Theorem 1.1 to $w(y) = u(x_0 + R_0 y)$. Dilation preserves the position of an eigenvalue in the Neumann spectrum and multiplies it by R_0^2 in this change of variables. Moreover, $\nabla_y w = R_0 \nabla_x u$ and $\operatorname{dist}(x, \partial\Omega) = R_0 \operatorname{dist}(y, \partial\Omega_\rho)$. These identities give (4.4); the critical-point statements are invariant under this change of variables. □

Remark 4.4 (Scope of the conclusion)*.* The neighborhood in Theorem 1.1 is measured in $C^{2,\alpha}$, with n and α fixed. The theorem makes no claim for an arbitrary convex domain in $\mathbb{R}^3$ or $\mathbb{R}^4$, nor for perturbations small only in Hausdorff distance or volume. It is consistent with convex counterexamples in sufficiently high dimensions, because its admissible neighborhood may depend on the dimension. The exact ball is included by taking $\rho = 0$.

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